Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 13 PDE

discuss in class

1. Show that the solution obtained by the Method of Separation of Variables to the PDE

$$u_x = t\ u_t$$

is of the form

$$u(x,t) = Ke^{\lambda x}\ t^{\lambda},$$

where K and λ are constants.

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2. Solve

$$u_t = 2u_{xx} \qquad \text{for} \quad 0 < x < 3, \quad t > 0$$

given that

$$u(0,t) = 0, \qquad u(3,t) = 0,$$

$$u(x,0) = 5\sin(4\pi x) - 3\sin(8\pi x) + 2\sin(10\pi x),$$

$$|u(x,t)| < M \quad \text{(i.e. } u \text{ is bounded) for } 0 < x < 3, \quad t > 0.$$

(a) Convert the above PDE problem into a Separation of Variable formulation and find the corresponding boundary and initial conditions.

(b) Solve for $u(x,t)$.

3. Using Laplace transform, solve the following wave equation

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \qquad 0 < x < 1, \quad t > 0$$

subject to the boundary conditions

$$u(0,t) = 0, \qquad u(1,t) = 0$$

and the initial conditions

$$u(x,0) = 0, \qquad \frac{\partial u}{\partial t}(x,0) = \sin \pi x, \quad 0 < x < 1.$$

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4. The partial differential equation given by

$$y_x - x^2 y_t = 0$$

with initial and boundary conditions given by

$$y(x,0) = 0 \quad \text{and} \quad y(0,t) = u(t), \text{ the unit step.}$$

(a) Solve for $y(x,t)$ by the method of Laplace Transform.

(b) Why do you think this partial differential equation can be solved by Laplace Transform technique?

(c) Why it does not help to apply Laplace Transform to solve

$$ty_x - xy_t = 0 \ ?$$

[Hint : Look up the property of differentiation in the Laplace Domain]

Answers

2. (b) $u(x,t) = 5e^{-32\pi^2 t} \sin 4\pi x - 3e^{-128\pi^2 t} \sin 8\pi x + 2e^{-200\pi^2 t} \sin 10\pi x$

3. $u(x,t) = \frac{1}{\pi}(\sin \pi t)(\sin \pi x)$

4. (a) $y(x,t) = u\left(t + \frac{x^3}{3}\right)$